

SIMATS ENGINEERING

CO3- ASSESSMENT TOOL 2

SIMULATION TASK

COURSE NAME: DIGITAL SIGNAL PROCESSING

COURSE CODE: ECA09

6. Truncation vs Rounding Error Spectrum

Simulate the frequency-domain characteristics of quantization errors produced by truncation

and rounding in a 20-tap FIR filter and compare their error spectra. (25) (BL5) (WK4)

Given Data:

- FIR Filter Length = 20 taps
- Input Signal = White Gaussian Noise
- Quantization = 8 bits
- FFT Length = 1024

Tasks:

1. Implement truncation-based quantization.
2. Implement rounding-based quantization.
3. Obtain error signals.
4. Compute FFT of error signals.
5. Plot error spectra.
6. Compare spectral characteristics.

AIM

To simulate and compare the frequency-domain characteristics of quantization errors produced by **truncation** and **rounding** in a **20-tap FIR filter** using an 8-bit quantizer and a 1024-point FFT.

OBJECTIVE

1. To design a 20-tap FIR filter and apply it to white Gaussian noise.
2. To implement **8-bit truncation quantization**.
3. To implement **8-bit rounding quantization**.
4. To obtain the quantization error signals for both methods.
5. To compute the FFT and error spectra using a **1024-point FFT**.
6. To compare the spectral characteristics of truncation and rounding errors.

TOOL USED

MATLAB — for FIR filter simulation, quantization, FFT computation, and plotting of error spectra.

MATLAB Code:

```
clc; clear; close all;
```

```
%% Parameters
```

```
N = 20;           % FIR filter length
```

```
bits = 8;         % Quantization bits
```

```
L = 1024;         % FFT length
```

```
%% Generate White Gaussian Noise
```

```
x = randn(1, L);
```

```
%% FIR Filter (Random coefficients)
```

```
h = randn(1, N);
```

```
%% Filter Output (Original)
```

```
y = filter(h, 1, x);
```

```
%% Quantization Step Size
```

```
q = 2^(-bits);
```

```
%% --- Truncation Quantization ---
```

```
y_trunc = floor(y / q) * q;
```

```
%% --- Rounding Quantization ---
```

```
y_round = round(y / q) * q;
```

```
%% Error Signals
```

```
e_trunc = y - y_trunc;
```

```
e_round = y - y_round;
```

```
%% FFT of Errors
```

```
E_trunc = fft(e_trunc, L);
```

```
E_round = fft(e_round, L);
```

```
%% Magnitude Spectrum
```

```
E_trunc_mag = abs(E_trunc);
```

```
E_round_mag = abs(E_round);
```

```
%% Frequency Axis
```

```
f = linspace(0, 1, L);
```

```
%% Plot
```

```
figure;
```

```
subplot(2,1,1);  
plot(f, 20*log10(E_trunc_mag));  
title('Error Spectrum (Truncation)');  
xlabel('Normalized Frequency');  
ylabel('Magnitude (dB)');  
grid on;
```

```
subplot(2,1,2);  
plot(f, 20*log10(E_round_mag));  
title('Error Spectrum (Rounding)');  
xlabel('Normalized Frequency');  
ylabel('Magnitude (dB)');  
grid on;
```

Output :

